LESSON PLAN

A/C MATERIAL (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	13
Total No. of Lesson Plans	16	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Control surface and tabs, aerodynamics characteristics of trailing edge controls, types of tabs, all moving surface, mechanical and structural aspects of control surface design
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Control surface and tabs, aerodynamics characteristics of trailing edge controls, types of tabs, all moving surface, mechanical and structural aspects of control surface design
5. Specific Objectives	15.1.1.1 Trainee will be able to explain aircraft structure, various structural elements and their details.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of control surface structural elements using IB diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary structural elements of an aircraft wing?
(b) Define the function of a trailing edge control surface.
(c) List the types of tabs used in aircraft control systems.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and function of control surfaces • Aerodynamic characteristics of trailing edge controls • Types of tabs and their operational mechanisms • All-moving surface design principles • Mechanical linkages and actuation systems	L & Ds	PPT, IB	15 min
(b)	• Structural elements of control surface design • Material selection and stress distribution • Hinge mechanisms and pivot points • Balance weights and aerodynamic stability • Corrosion protection and maintenance access	L & Ds	PPT, PDF	20 min
(c)	• Integration of mechanical and structural aspects • Failure modes and structural integrity checks • Inspection criteria for control surface hardware • Impact of structural distortion on control effectiveness • Summary of design standards and specifications	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a tab modifies the aerodynamic force on a control surface.
(b) Describe the structural role of balance weights in an all-moving surface.
(c) Identify two common failure modes in control surface hinge mechanisms.

12. Home Assignment and Follow Up	Prepare a brief note on the structural implications of aero-elasticity on control surface design, referencing the concepts introduced in the next lesson plan.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to aero-elasticity, classification, effects of structural distortion, wing torsional divergence, control effective and reversal. introduction to flutter, control surface flutter, wing bending torsional flutter, factor effecting flutter characteristics, aileron buzz, buffeting, dynamic response, effect of sweep on various critical speeds	Refer to the master syllabus reference list	LP 14

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Verify specific reference page numbers for control surface structural diagrams in the current IB edition.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min